

Classification ML Model

Predictive Analysis for Customer Retention in Brazilian E-Commerce

By:

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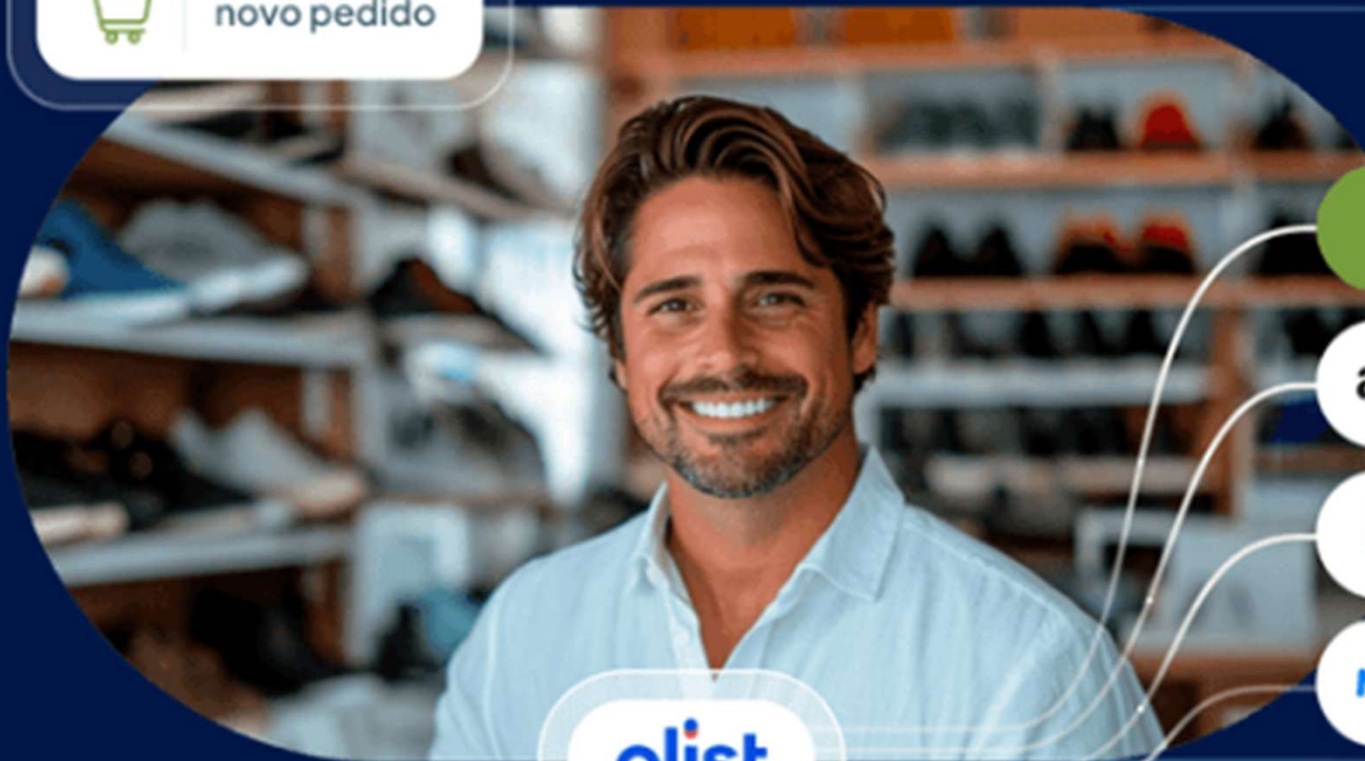
- 📌 Business Understanding
- 📌 Data Understanding
- 📌 Data Preparation
- 📌 Modeling
- 📌 Conclusions
- 📌 Business Recommendations



Business Understanding



novο pedido



olist



amazon

americanas

magalu

Problem Definition

Predictability of customer loyalty to the e-commerce

Project Objective

Making predictions about customer repurchase trends using ML

Marketplace Benchmarking

- ¹ Customers who make repeat purchases buy 67% more than first-time buyers.
- ¹ For marketplace sellers, the repeat purchase ratio is 20-40%
- ² The retention rate for Walmart and Target over 16 months is 14%, for Temu it's 28%, and for Amazon it's 56%.

Sources:

¹ Salesduo, ² Earnest Analytics

Marketplace Benchmarking

- ³ Approximately 90% of transactions involve disintermediation after the first purchase.
- ⁴ Less priority on delivery speed and more on delivery reliability.
- ⁴ Importance of free shipping and discounts, and retention programs.

Sources:

³ Zhu et al. 2018, ⁴ McKinsey

⁵ Mercado Livre: Brazilian Marketplace Example

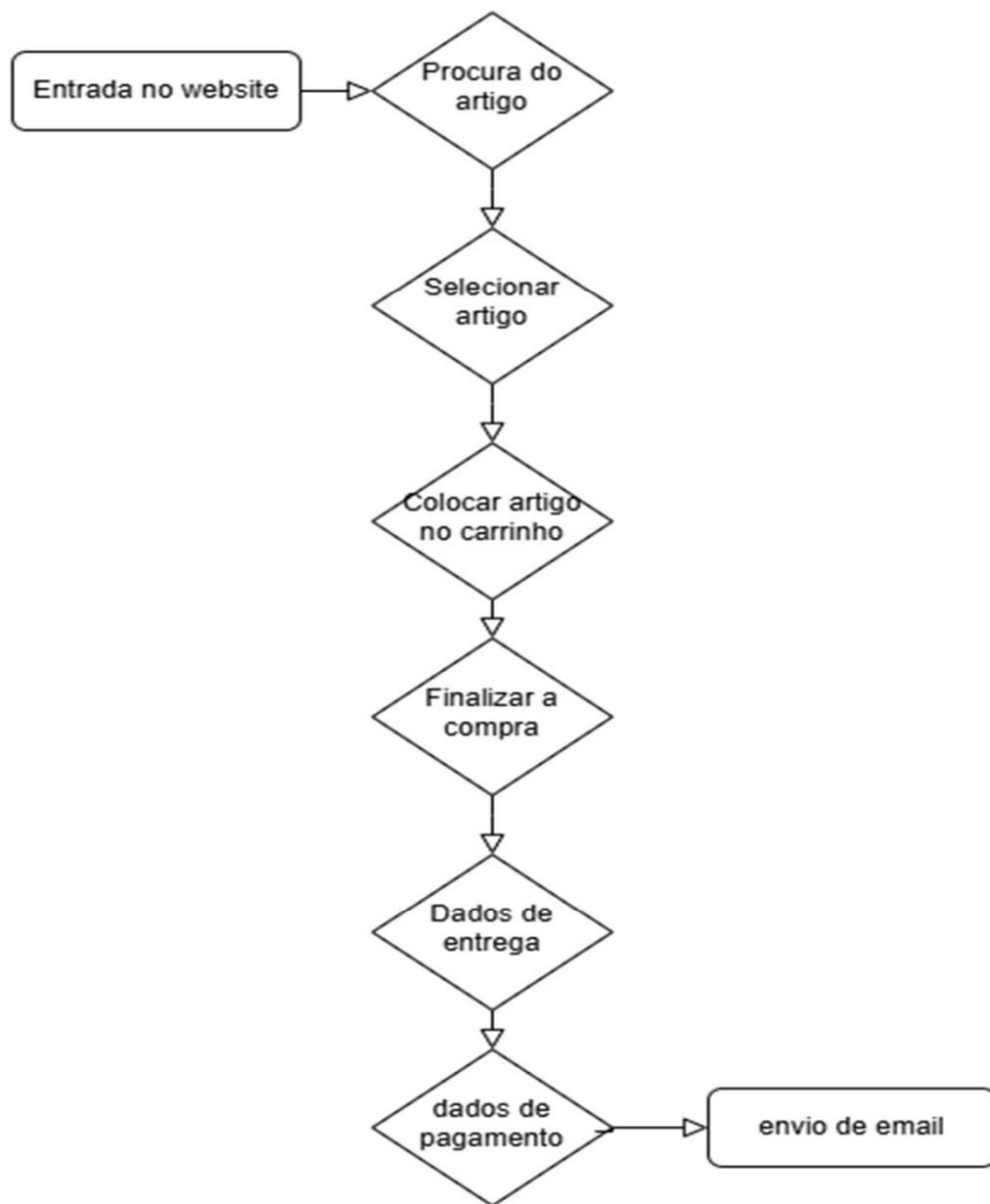
Good coupon policy.

Good customer communication policy.

Good free shipping policy.

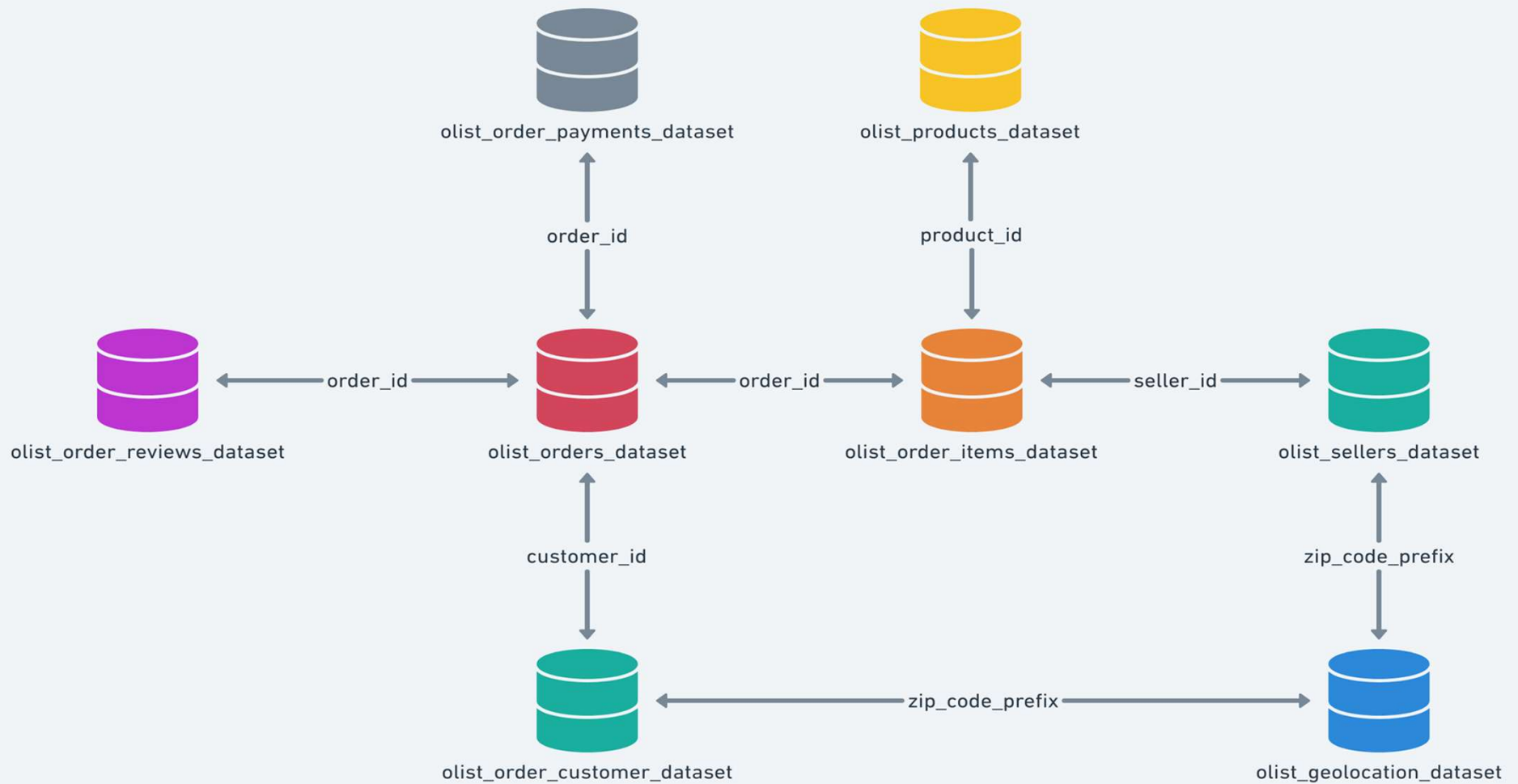
Sources:

⁵ Mercado Livre



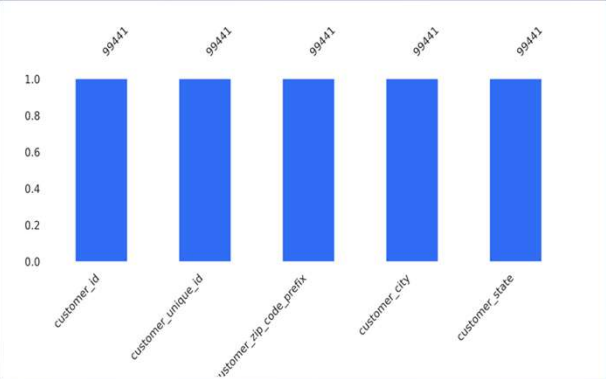
Data Understanding

Olist Database Schema



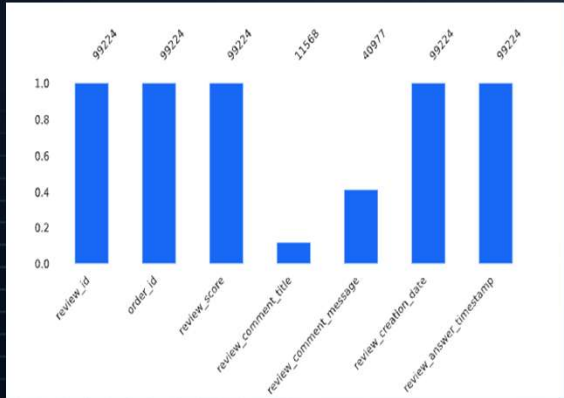
Tables Selection

Customers



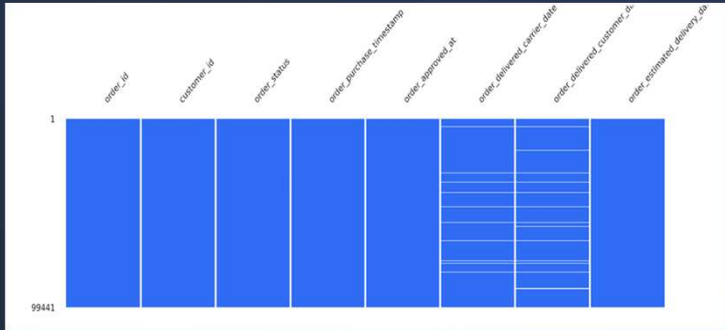
No Missing Values

Reviews



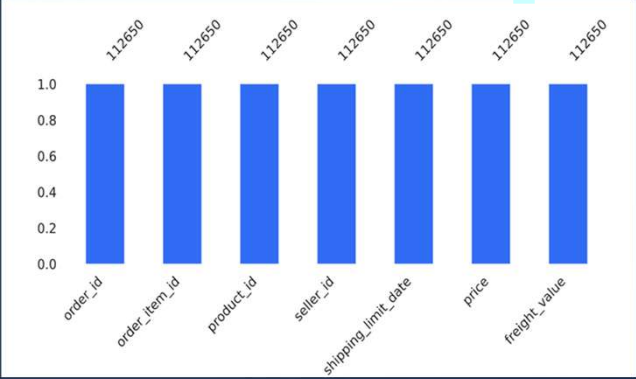
review_comment_title 88.3% missing values
review_comment_message 58.7% missing values

Orders



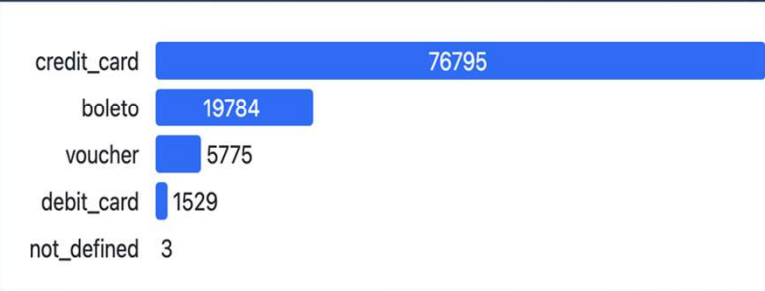
Order_delivered_carrier_date 1.8% missing values
Order_delivered_customer_date 3.0% missing values

Orders Items



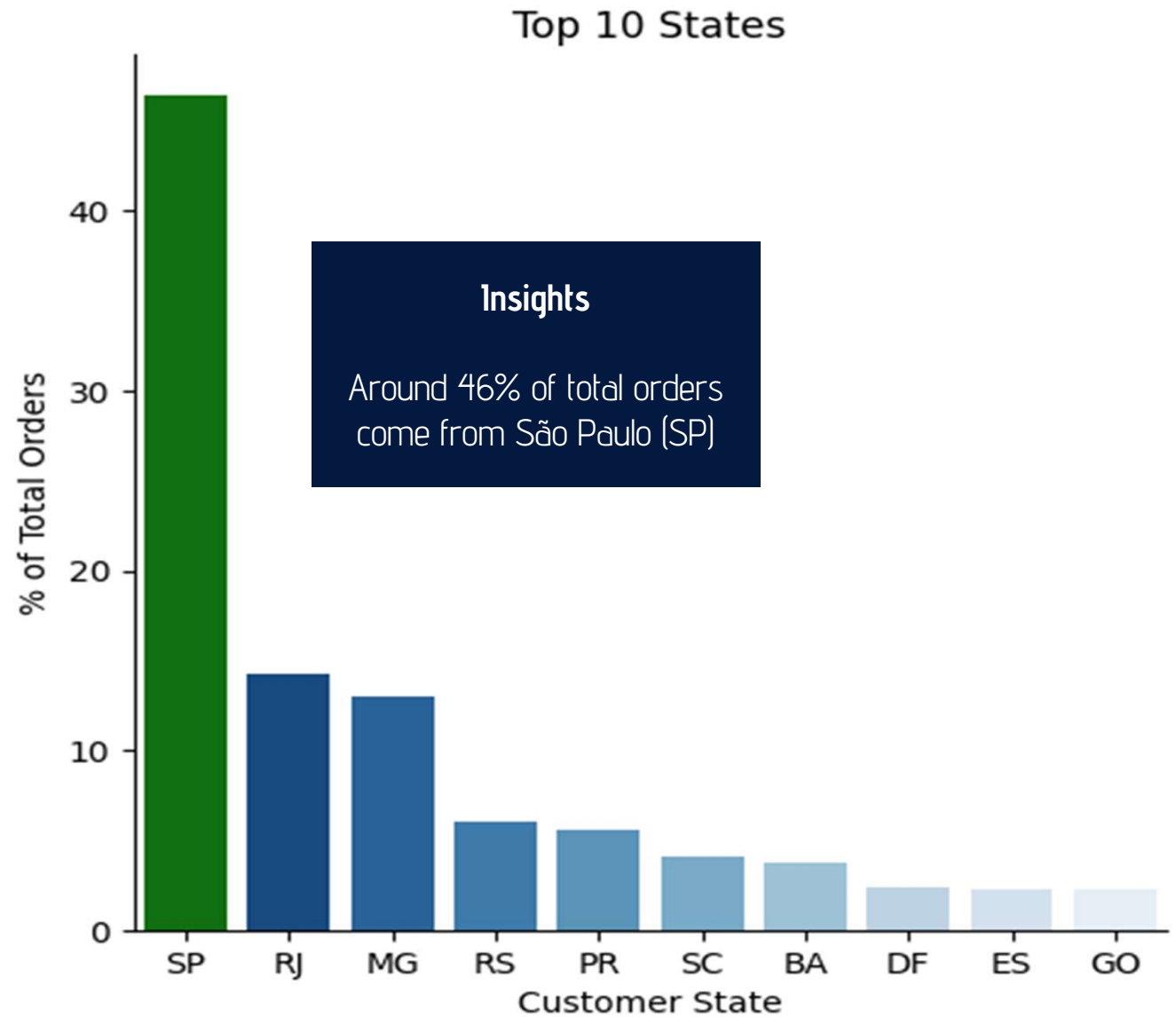
No Missing Values

Payments

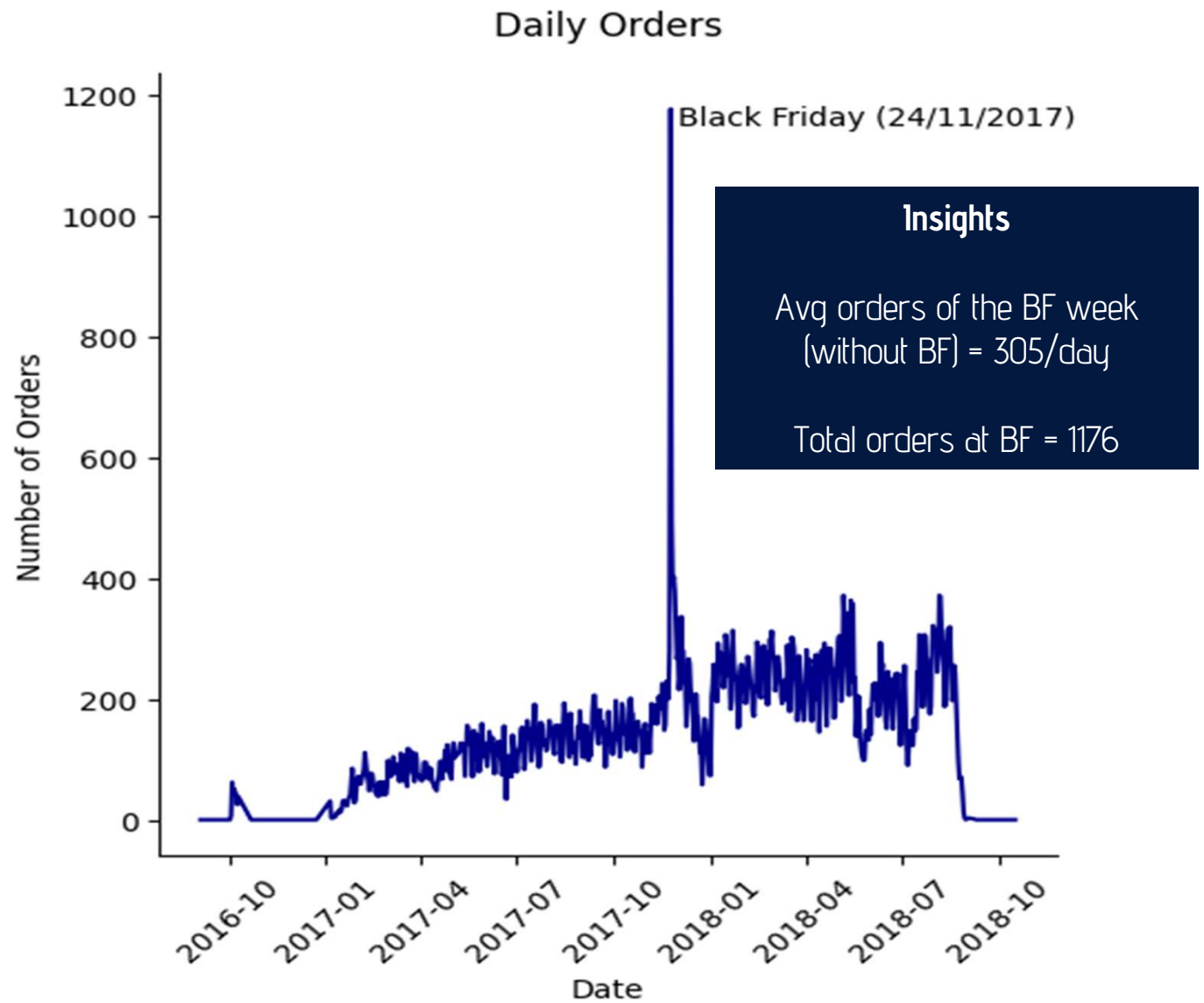


No Missing Values

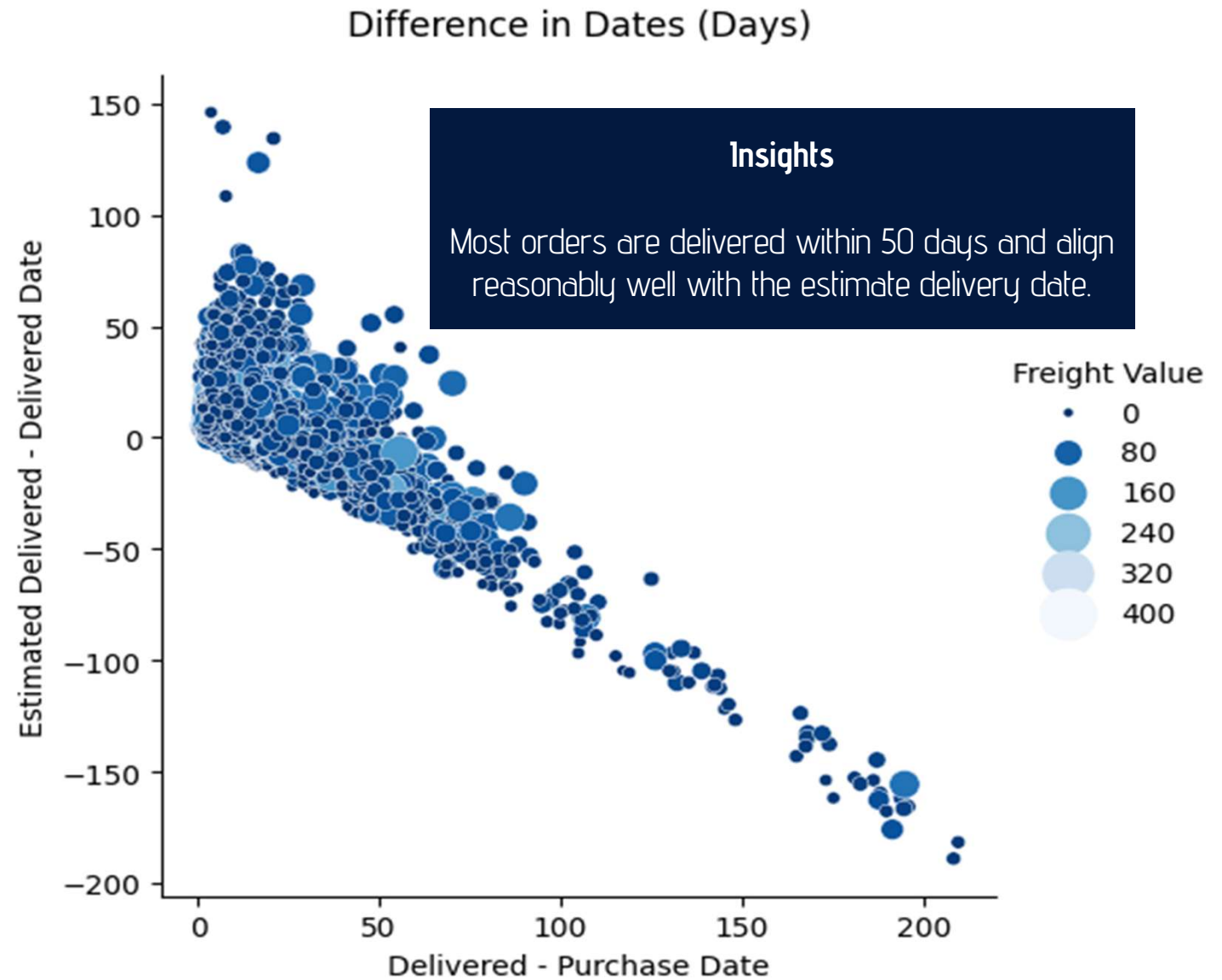
Initial EDA: Important Insights



Initial EDA: Important Insights



Initial EDA: Important Insights

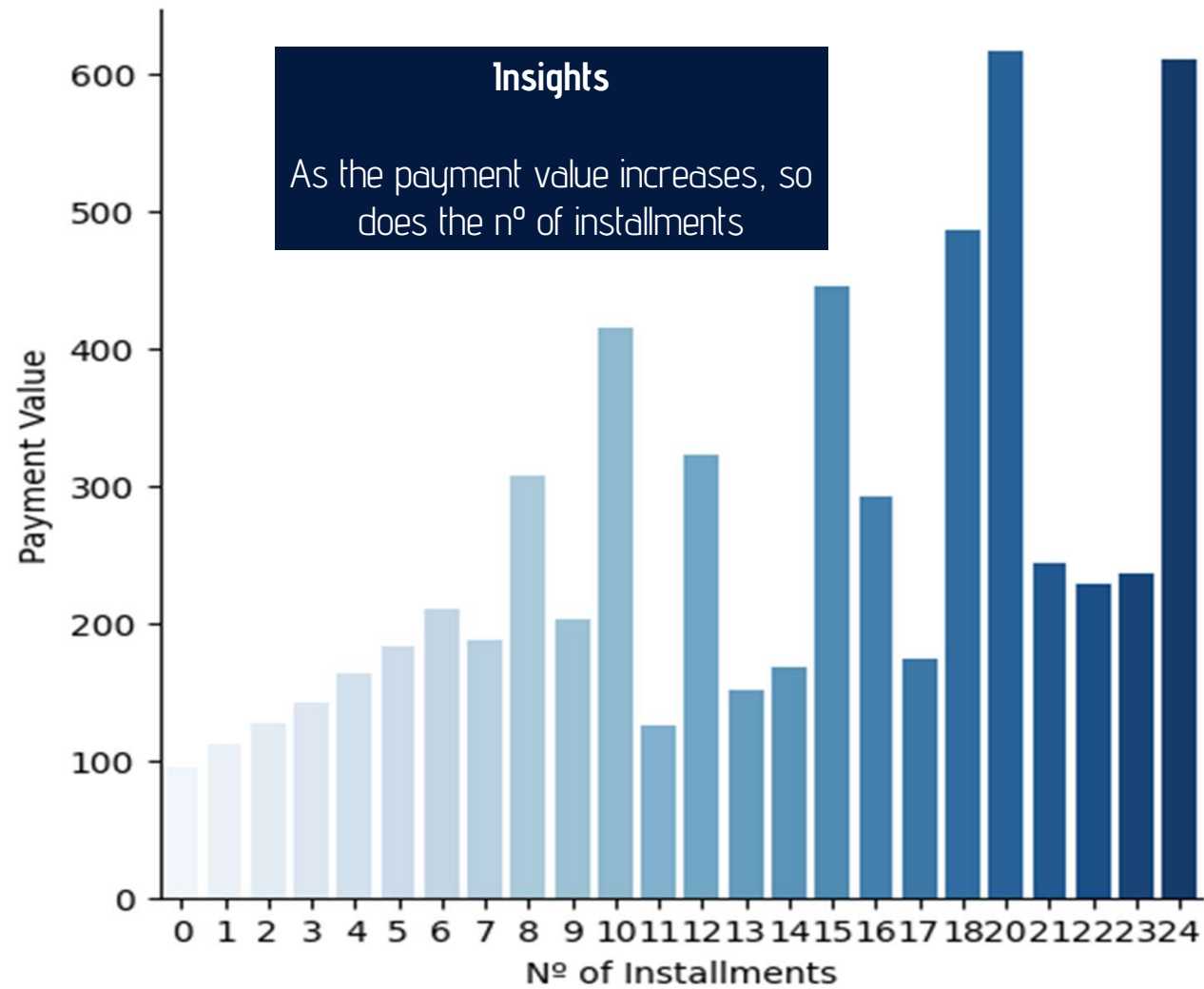


Initial EDA: Important Insights



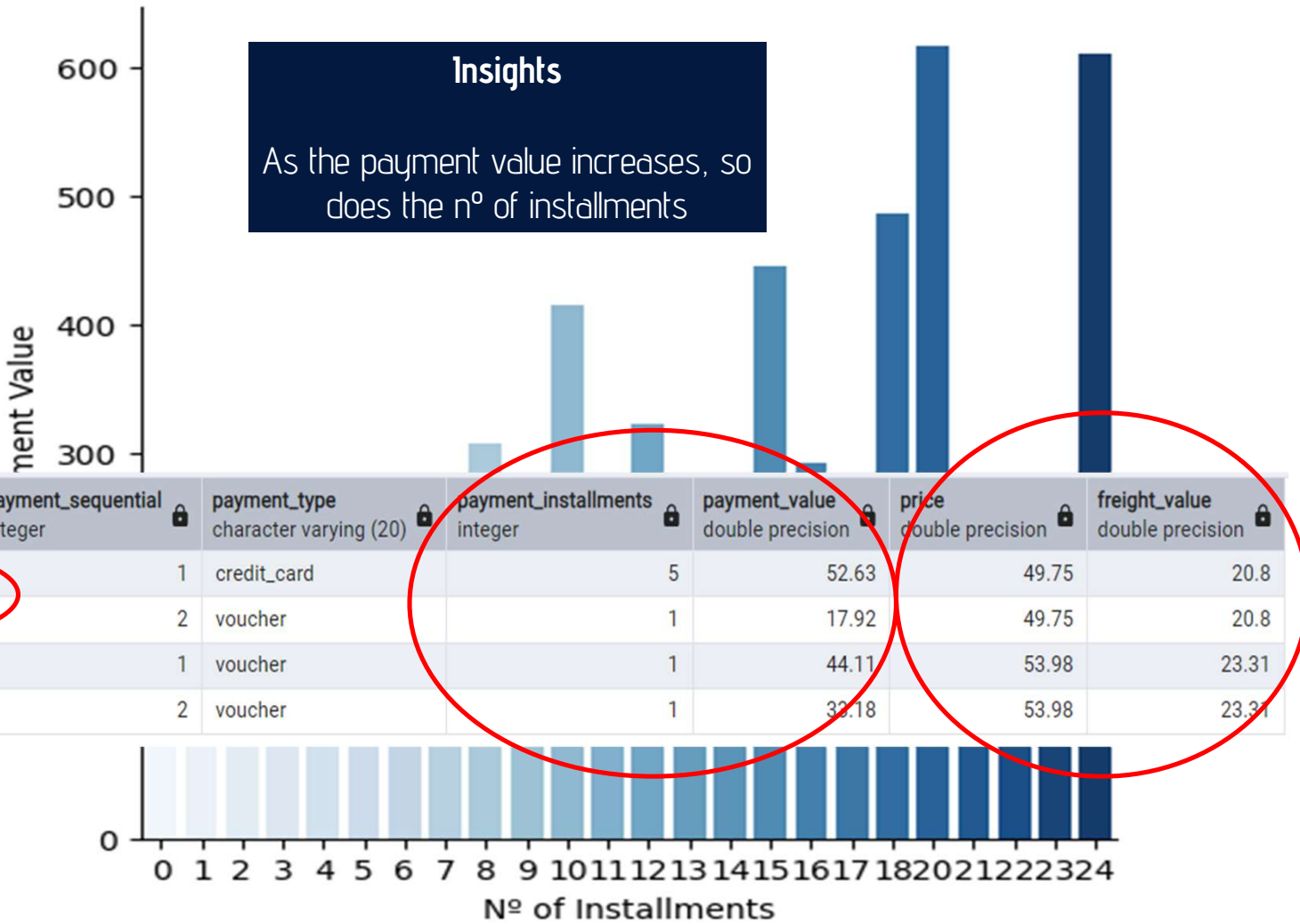
Initial EDA: Important Insights

Payment Value by N° of Installments



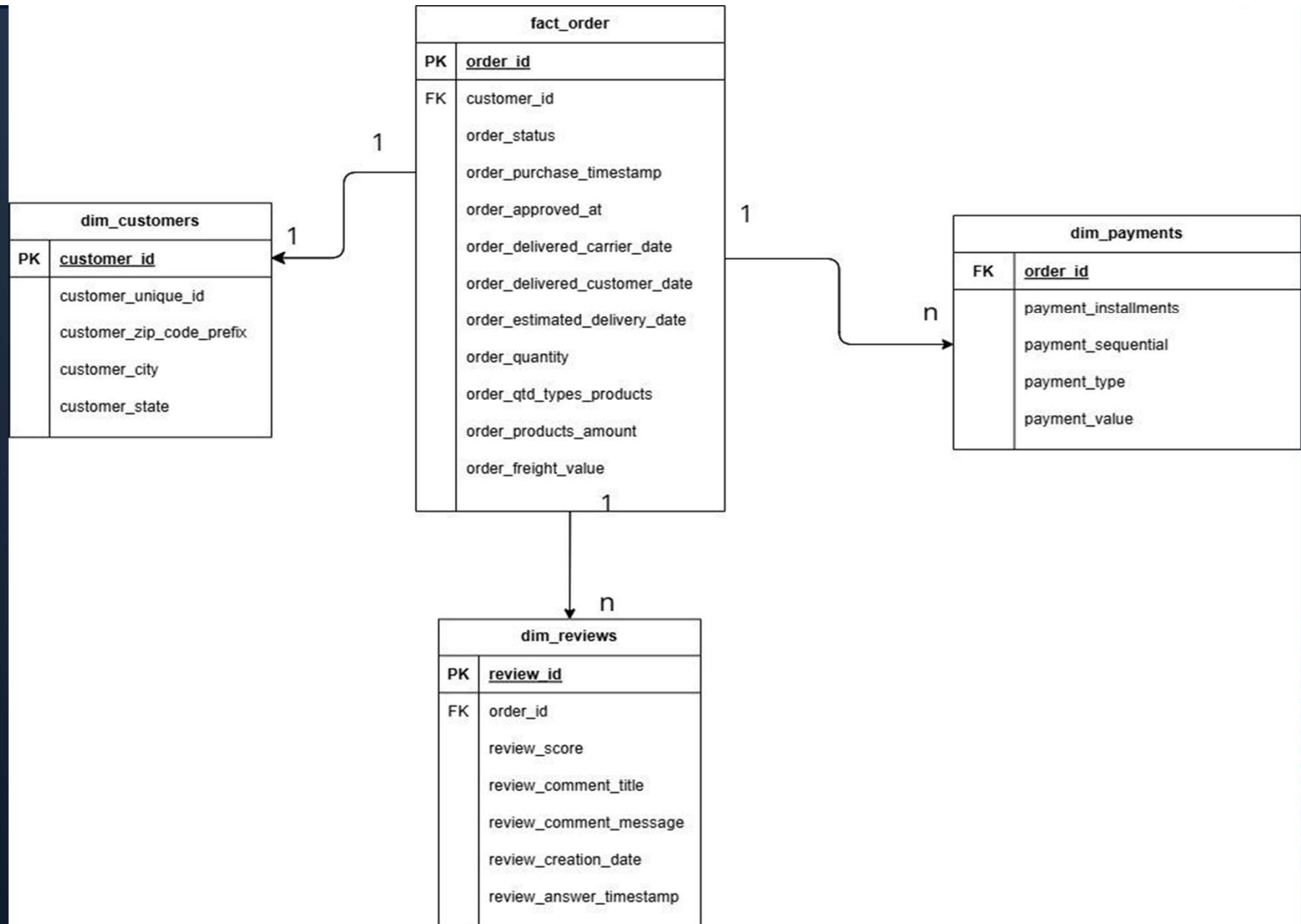
Initial EDA:
Important
Insights

Payment Value by N° of Installments

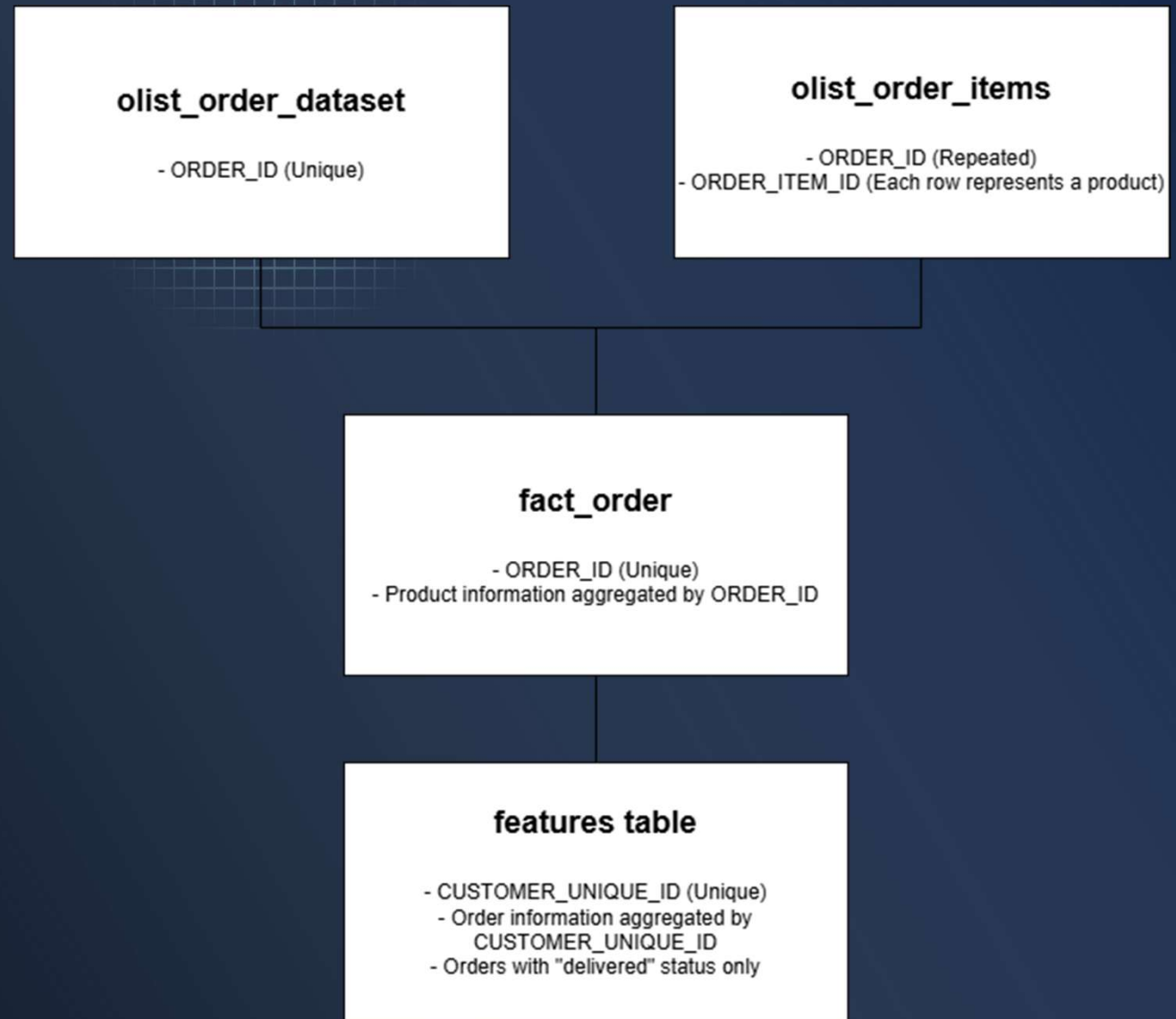


Data Preparation

Star Schema Building



Data Hierarchization



TARGET

1- A temporal separation was made in the data, in order to create a 'past' moment and a 'future' moment.

2- We chose the date 05/31/2017 to be the dividing point

3 - We created an SQL query that builds the predict_next_order_delivered table.



4 - The granularity of this table is each customer in the Olist database who made a purchase up to 05/31/2017.

5 - The TARGET variable ('retention') receives the value 0 if the customer who purchased up to 05/31/2017 did not buy again after that date, and the value 1 if they did buy again.

6 - The remaining columns are the features that describe the behavior of these customers.

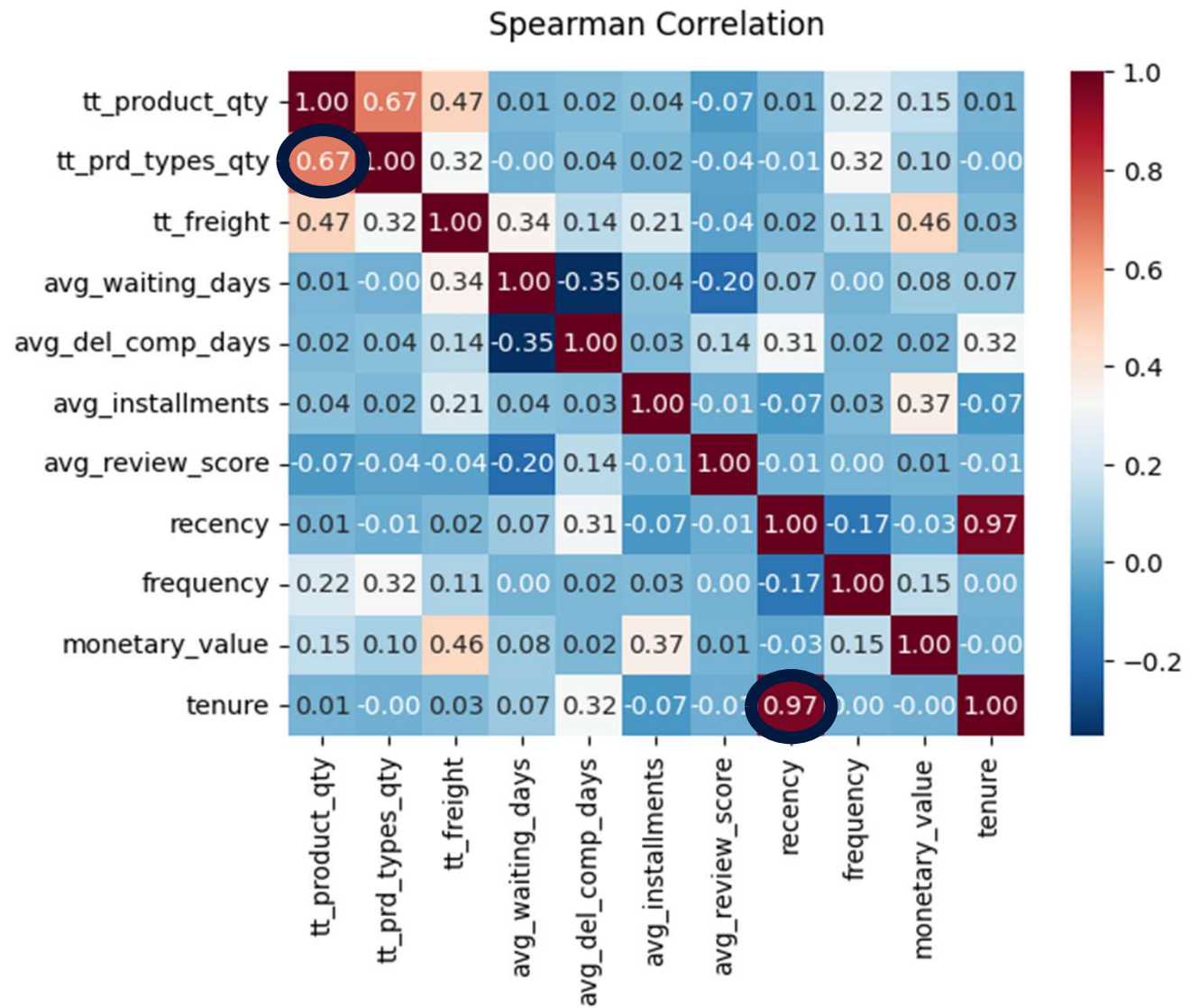
Note: The first order was placed on 09/04/2016 and the last one on 10/17/2018.

Features Table Preview

Customer	Retention	Products	Types of Products	Freight	Avg Waiting Days	Avg Delivery Compliance Days	Avg Installments	Avg_Review	Day/Night	Week/Weekends	Payment Type	State
client_123	1	5	3	35.00	7	2	2	4.5	day	weekdays	credit_card	sp_rj_mg
client_456	0	2	2	18.50	10	1	1	3.0	night	weekend	boleto	other_state
client_789	1	1	1	12.00	5	3	3	5.0	day	weekdays	credit_card	sp_rj_mg
client_abc	0	3	2	25.00	8	0	1	4.0	day	weekdays	debit_card	other_state

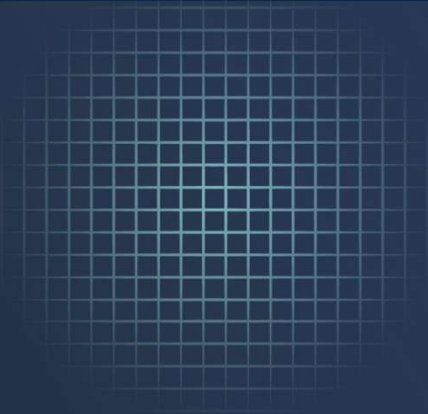
No retention: 97%
Retention: 3%

Correlation for Feature Selection



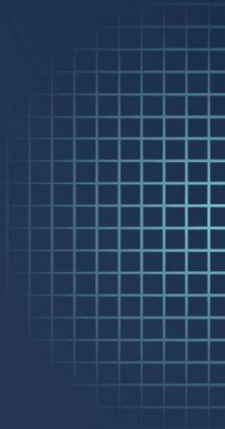


Modelling



Clustering

Customer Segmentation



K-Means Methodology

Feature Selection

- Various feature combinations across multiple iterations
- Features related to customer characteristics, such as location and spending habits

Data Preprocessing

- One-hot encoding
- Scaling numerical features to the 0-1 range

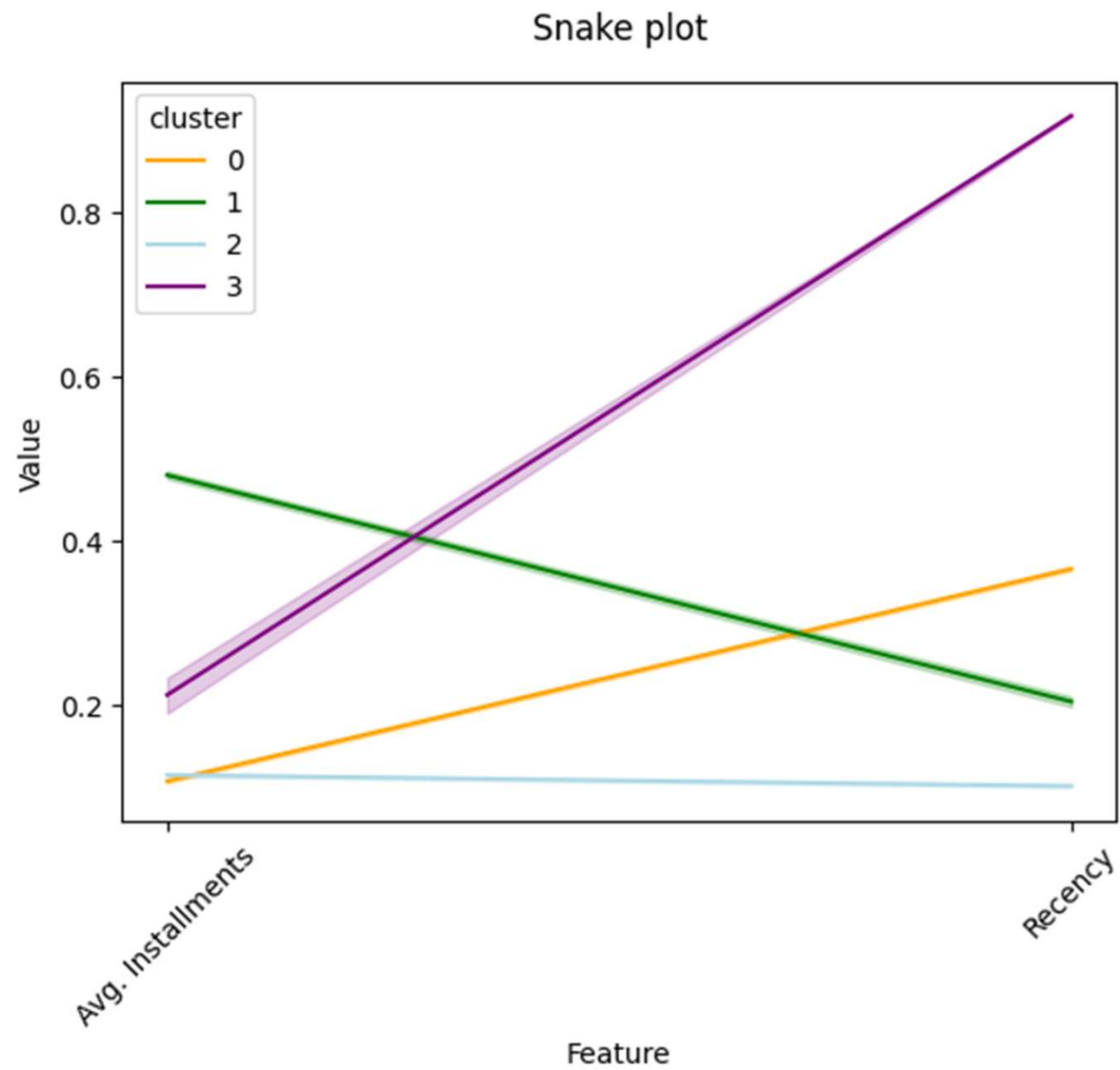
Cluster Analysis

- Elbow method and Silhouette score

Challenges

- High cardinality in categorical variables (K-Modes for future reference)
- Monetary value distribution: Highly skewed

Cluster Analysis



Cluster Analysis

Reactivation-Ready Spenders

Cluster 0

Characteristics:

- Recent buyers but not brand new

Marketing Strategy:

- Re-Engagement campaigns

Premium Installment Spenders

Cluster 1

Characteristics:

- High Spending
- Comfortable with spreading out payments

Marketing Strategy:

- Offer flexible payment options
- Premium products

New Value Spenders

Cluster 2

Characteristics:

- Largest segment
- Recent purchases

Marketing Strategy:

- Onboarding and Welcome campaigns

Long-Lapsed Spenders

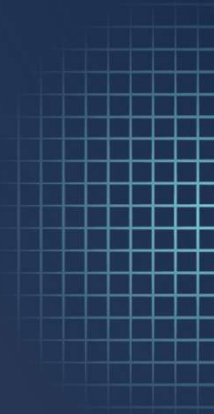
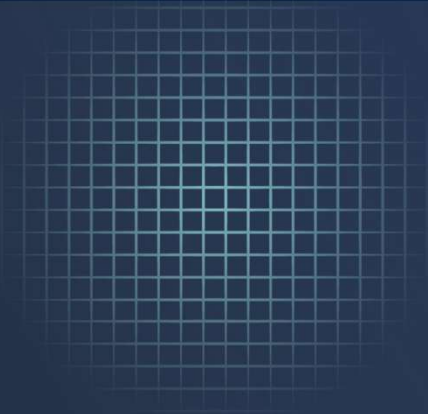
Cluster 3

Characteristics:

- Smallest segment
- Low spending
- Long inactivity

Marketing Strategy:

- Evaluate if the effort to re-engage them is cost-effective



Classification

Random Forest Methodology

Feature Selection

- Various feature combinations across multiple iterations
- Features that could impact customer's likelihood to make a repeat purchase

Data Preprocessing

- One-hot encoding
- Numerical imputation for average score feature
- Pipeline to avoid data leakage

Model Training

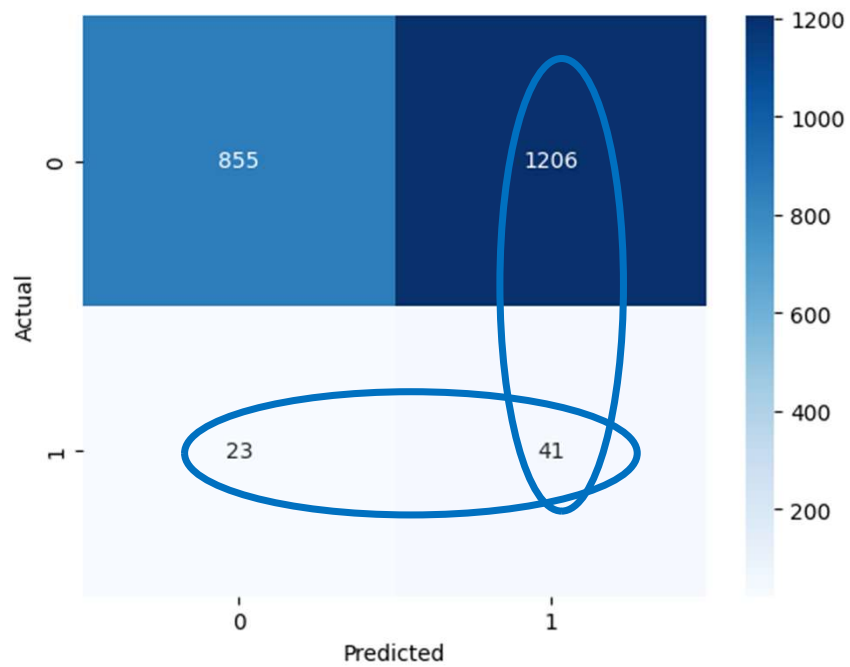
- Train/ Test split
- RandomSearchCV (K-fold = 10) trained on the training dataset with Recall score and then evaluated on the independent test dataset

Challenges

- Introducing bias through mean and median imputation (Other imputation for future reference)
- Imbalanced class

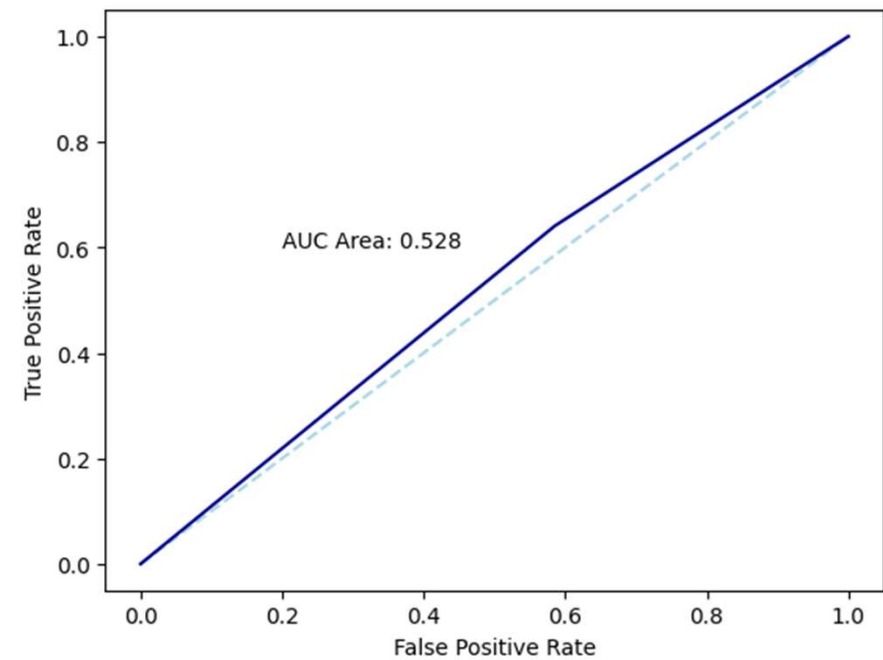
RF Retention Analysis

Confusion Matrix



Train Recall	Test Recall
0.68	0.64

Random Forest ROC Curve



Test Set	Recall	Precision
0	0.41	0.97
1	0.64	0.03

Retention Analysis

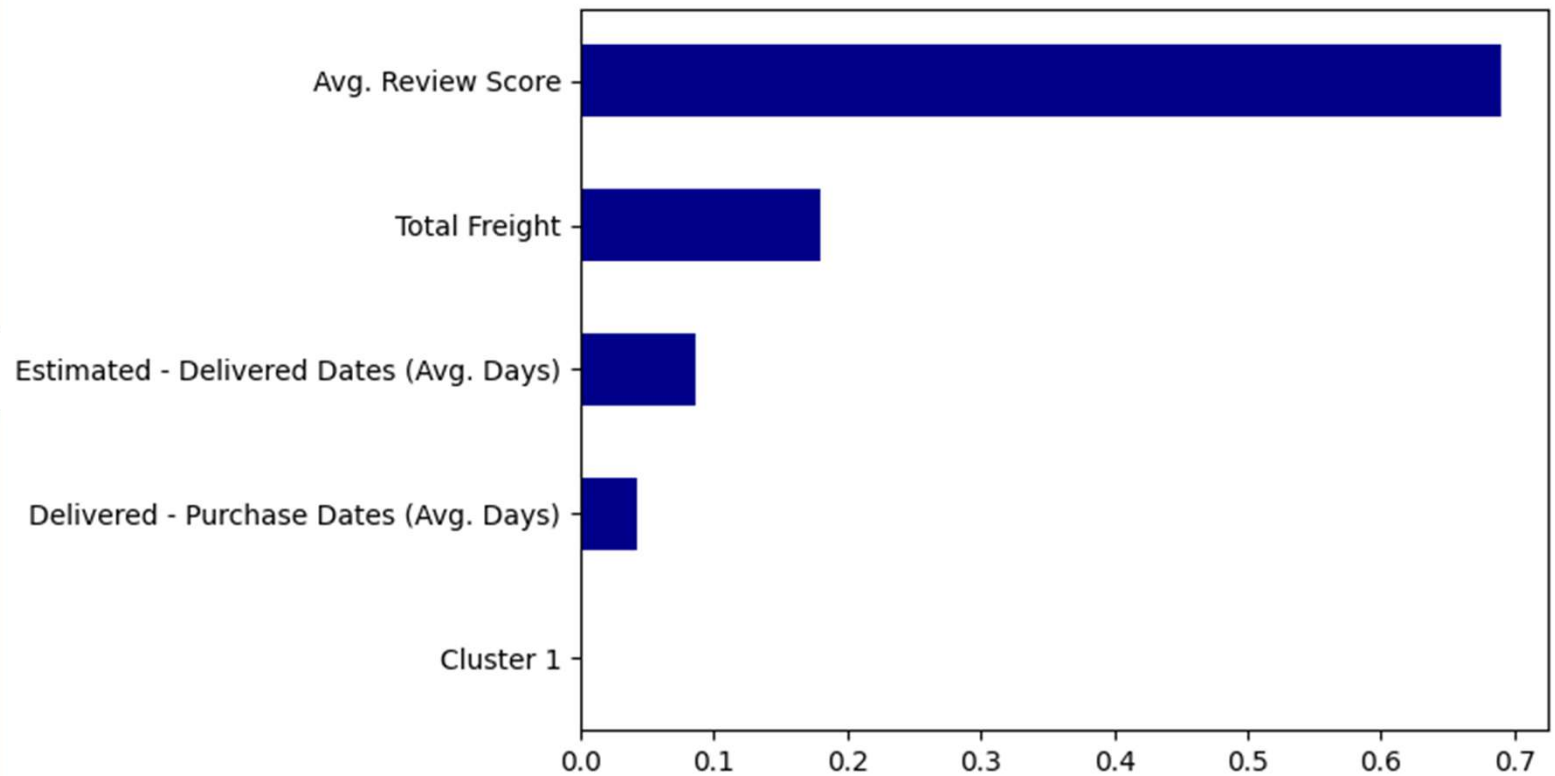
Relevant Factors

Product/ Service quality

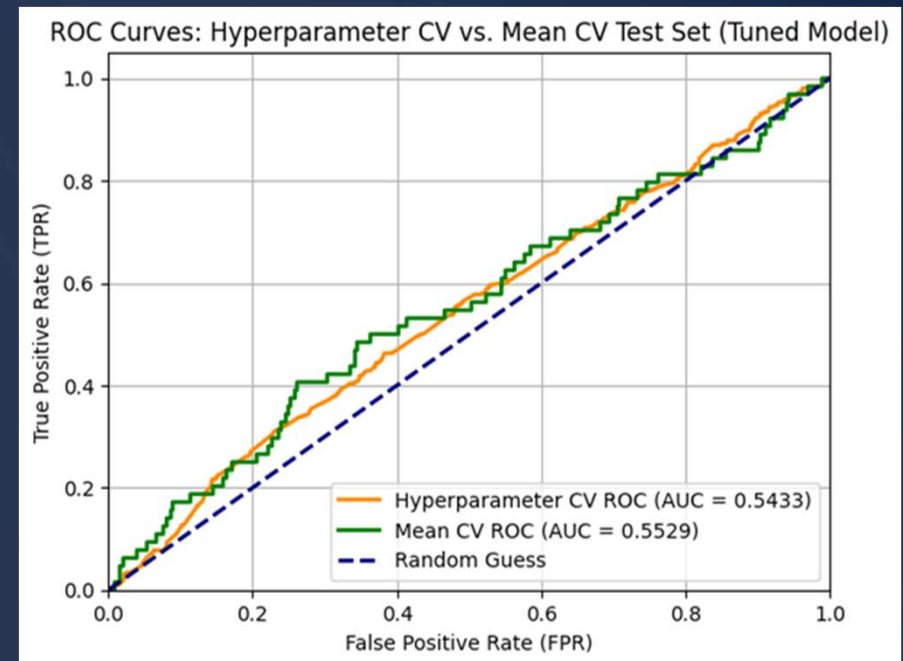
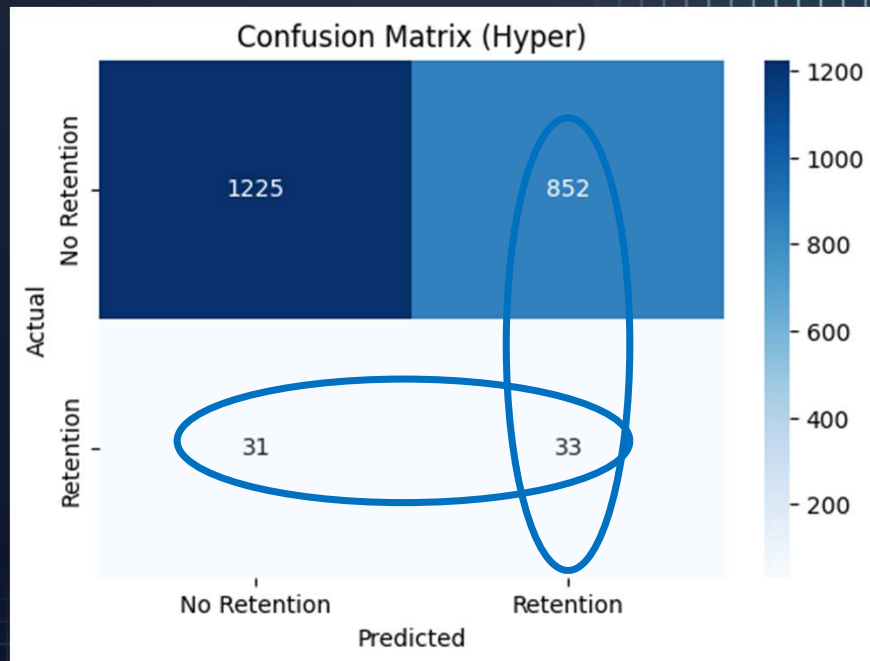
Logistic issues

The model's performance overview is poor due to imbalanced data

Feature Importance



LR - Retention Analysis



Test Set	Recall	Precision
0	0.59	0.98
1	0.52	0.04



Conclusions

Conclusion: Key Takeaways



Imbalanced Data Impact

The dataset's imbalance caused the model to favor the majority class, affecting its performance.



Missing Data Challenges

Absence of data limited the model's ability to learn comprehensive patterns, contributing to performance constraints.



Best Performance Model

Random Forest performed better than Logistic Regression for this case scenario.



+ Retention Data

To predict the next order more accurately, more retention data was needed.

Conclusion: The diamond customers

True Positive = 41 customers

The Random Forest model detected correctly 41 customers that purchased again.

These customers are very important because they represent direct revenue to the business and must be nurtured by the marketing department.

Conclusion: Possible hidden opportunities

False Positive = 1206 customers

The Random Forest model detected that 1206 customers purchased again, but actually they didn't.

Even though this detection is incorrect, it's important to conduct a cost analysis in order to potentially run campaigns to attract these customers.

Conclusion: Black Friday Insights

Orders at BF = 1176

Number of customers that purchased before the time split (31/05/2017) and purchased again at BF

=

2 Customers

Based on the numbers above, it is possible to see that Black Friday is an excellent tool for a massive injection of revenue at one time.

However, it did not prove to be a good tool for increasing customer retention.



Business Recommendations

**The client needs to increase
the customer retention rate
in order to have a
Classification Model with
better performance metrics.**

Some recommendations based on client data



Customer Segments

- Use the information from the clusters to apply marketing campaigns adapted to each type of customer.



Logistic improvements

- Work with sellers to improve logistic KPIs (On-time delivery rate)
- Create “free freight” campaigns



Loyalty programs

- Create loyalty programs to prevent customers from bypassing the marketplace and buying directly from the Seller

Bibliographic References

¹ <https://salesduo.com/blog/customer-retention-on-amazon/>

² <https://www.earnestanalytics.com/insights/temus-retention-grows-over-time-leads-walmart-trails-amazon>

³ https://mackinstitute.wharton.upenn.edu/wp-content/uploads/2021/03/Gu-Grace_Technology-and-Disintermediation-in-Online-Marketplaces.pdf

⁴ <https://www.mckinsey.com/industries/logistics/our-insights/what-do-us-consumers-want-from-e-commerce-deliveries>

⁵ <https://www.mercadolivre.com.br/l/promocoes>

Thank you!

Do you have any questions?